

A Dyadic-Block Discrete Schrödinger Counterexample to the Density-of-States Converse

Candidate result produced in run `fa_banach_001`, lane 2

9 August 2026

Status. Candidate full negative answer to the *discrete* Schrödinger question in arXiv:2202.03676; likely valid, pending expert review. The continuous Euclidean question is not addressed.

1 Source question

Azamov, Hekkelman, McDonald, Sukochev, and Zanin construct a bounded self-adjoint diagonal operator with no density of states (DOS), although $f(H)M_w$ is Dixmier measurable for every $f \in C_c(\mathbb{R})$. They then ask whether this separation is still possible for Schrödinger operators [1, p. 6]:

“However, the above example is not a Schrödinger operator. It is unknown whether these definitions of the DOS still differ when restricting to Schrödinger operators.”

The source page is reproduced below.

6

N. AZAMOV, E. HEKKELMAN, E. MCDONALD, F. SUKOCHEV, D. ZANIN

Hence,

$$\frac{1}{\log(2+n)} \sum_{k=1}^n \lambda(k, f(H)M_w) \rightarrow \frac{f(0) + f(1)}{2},$$

and by [33, Theorem 9.1.2(c)] this means that $f(H)M_w$ is Dixmier measurable.

What we can learn from this example is that requiring $f(H)M_w$ to be Dixmier measurable for all $f \in C_c(\mathbb{R})$ is a strictly weaker assumption on self-adjoint operators H than requiring the existence of a DOS. Hence $\text{Tr}_w(f(H)M_w)$ can be interpreted as a strictly stronger extension of the definition of the DOS on metric spaces that satisfy the conditions of Theorem 1.2.

However, the above example is not a Schrödinger operator. It is unknown whether these definitions of the DOS still differ when restricting to Schrödinger operators. Note that for the continuous case, where we take a Schrödinger operator $H = -\Delta + M_V$ on $L_2(\mathbb{R}^d)$, it is also still an open question how the existence of the DOS measure for H is related to the Dixmier measurability of $f(H)M_w$ for

The result below answers this discrete question affirmatively: the two conditions still differ even for a nearest-neighbor half-line Schrödinger operator with a bounded $\{0, 1\}$ -valued potential.

2 Definitions and construction

Let $X = \mathbb{N} = \{1, 2, \dots\}$ have its usual metric and base point $x_0 = 1$, and let $(e_n)_{n \geq 1}$ be the standard basis of $\ell_2(\mathbb{N})$. The Dirichlet discrete Laplacian is

$$(\Delta_D x)_n = 2x_n - x_{n-1} - x_{n+1}, \quad x_0 = 0.$$

It is a bounded positive operator with spectrum in $[0, 4]$.

Define $v : \mathbb{N} \rightarrow \{0, 1\}$ by $v_1 = 1$ and, for $m \geq 0$,

$$v_n = 0 \quad (2^{2m} < n \leq 2^{2m+1}),$$

while, for $m \geq 1$,

$$v_n = 1 \quad (2^{2m-1} < n \leq 2^{2m}).$$

Thus v is constant on successive dyadic blocks and alternates between zero and one. Put

$$H = \Delta_D + M_v, \quad W = M_w, \quad w_n = \frac{1}{n}.$$

Then H is a bounded discrete Schrödinger operator, $0 \leq H \leq 5$, and $W \in \mathcal{L}_{1,\infty}$.

For $c \in \{0, 1\}$, let $\Delta_{\mathbb{Z}}$ be the same Laplacian on $\ell_2(\mathbb{Z})$ and define the bulk state

$$\tau_c(f) = \langle \delta_0, f(\Delta_{\mathbb{Z}} + cI)\delta_0 \rangle, \quad f \in C([0, 5]).$$

3 Proof intuition

The construction keeps the source paper's alternating dyadic potential but adds genuine nearest-neighbor hopping. Ordinary volume averages see the *lengths* of the blocks. Since each new block is as long as everything before it, the fraction of sites with potential one oscillates between $1/3$ and $2/3$; already the test function equal to the identity on the spectrum prevents a DOS.

Dixmier averages use the logarithmic weight $1/n$. Every dyadic block then has asymptotic weight $\log 2$, so the zero and one blocks each receive half the mass. Nearest-neighbor hopping does not spoil this balance. For a polynomial p , the diagonal entry $\langle e_n, p(H)e_n \rangle$ only sees a fixed neighborhood of n . Away from the dyadic interfaces it is exactly the constant-potential bulk value $\tau_{v_n}(p)$; the total $1/n$ -weight of all fixed-width interface neighborhoods is finite. Uniform polynomial approximation extends the logarithmic limit to every continuous f . The source's modulated-operator eigenvalue formula then converts the diagonal limit into Dixmier measurability of $f(H)W$.

4 The counterexample theorem

Theorem 1. *For the operator $H = \Delta_D + M_v$ above:*

- (i) *H has no density of states with respect to the balls $\{1, \dots, N\}$;*
- (ii) *for every $f \in C_c(\mathbb{R})$, the operator $f(H)W$ is Dixmier measurable. More precisely, if $\lambda_k(f(H)W)$ is an eigenvalue sequence ordered by nonincreasing modulus, then*

$$\lim_{N \rightarrow \infty} \frac{1}{\log(2 + N)} \sum_{k=1}^N \lambda_k(f(H)W) = \frac{\tau_0(f) + \tau_1(f)}{2}.$$

Consequently, Dixmier measurability of $f(H)W$ for all $f \in C_c(\mathbb{R})$ does not imply existence of the DOS, even within bounded one-dimensional discrete Schrödinger operators.

4.1 Failure of the density of states

Let P_N be the projection onto $\text{span}\{e_1, \dots, e_N\}$. Direct summation of the dyadic blocks gives

$$\frac{1}{2^{2m}} \sum_{n=1}^{2^{2m}} v_n \longrightarrow \frac{2}{3}, \quad \frac{1}{2^{2m+1}} \sum_{n=1}^{2^{2m+1}} v_n \longrightarrow \frac{1}{3}.$$

Choose $g \in C_c(\mathbb{R})$ with $g(t) = t$ on $[0, 5]$. Since $\text{spec}(H) \subset [0, 5]$, functional calculus gives $g(H) = H$. Moreover, $\langle e_n, H e_n \rangle = 2 + v_n$, and therefore

$$\frac{1}{N} \text{Tr}(P_N g(H) P_N) = 2 + \frac{1}{N} \sum_{n=1}^N v_n.$$

This expression has two different subsequential limits. The defining DOS limit fails for the single test function g , proving part (i).

4.2 The logarithmic block balance

Lemma 1. *For $c \in \{0, 1\}$,*

$$A_c(N) := \sum_{\substack{1 \leq n \leq N \\ v_n = c}} \frac{1}{n} = \frac{1}{2} \log N + O(1).$$

Proof. For $j \geq 0$, the harmonic mass of the dyadic block $I_j = (2^j, 2^{j+1}] \cap \mathbb{N}$ is

$$h_j = \sum_{n=2^j+1}^{2^{j+1}} \frac{1}{n} = \log 2 + O(2^{-j}).$$

The potential is constant on I_j and alternates with the parity of j . The alternating sum of the complete block masses is bounded: the constant $\log 2$ terms cancel in pairs and the errors are summable. A final partial block has mass at most $h_j = O(1)$. Hence $A_1(N) - A_0(N) = O(1)$. Since $A_1(N) + A_0(N) = \sum_{n \leq N} n^{-1} = \log N + O(1)$, the assertion follows. $\square$

4.3 Locality across the interfaces

Lemma 2. *For every polynomial p ,*

$$\sum_{n=1}^N \frac{\langle e_n, p(H) e_n \rangle}{n} = \frac{\tau_0(p) + \tau_1(p)}{2} \log N + O_p(1).$$

Proof. Let $r = \deg p$. Because H has propagation one, the diagonal coefficient $\langle e_n, p(H) e_n \rangle$ depends only on the matrix of H in the radius- r neighborhood of n . If $n > r$ and that neighborhood meets no interface $\{2^j : j \geq 0\}$, the potential is constant there. The corresponding matrix coefficient is therefore exactly

$$\langle e_n, p(H) e_n \rangle = \tau_{v_n}(p).$$

Let E_r consist of $\{1, \dots, r\}$ together with the integers at distance at most r from some 2^j . Its harmonic mass is finite. Indeed, after discarding finitely many small j ,

$$\sum_{\substack{n \in \mathbb{N} \\ |n - 2^j| \leq r}} \frac{1}{n} \leq \frac{2(2r + 1)}{2^j},$$

and the right-hand side is summable in j . Since both H and the two bulk operators have spectrum in $[0, 5]$, the discrepancy at a bad site is bounded by $2\|p\|_{C([0,5])}$. Consequently,

$$\sum_{n=1}^N \frac{\langle e_n, p(H)e_n \rangle}{n} = \tau_0(p)A_0(N) + \tau_1(p)A_1(N) + O_p(1).$$

Lemma 1 finishes the proof. $\square$

Lemma 3. *For every $f \in C([0, 5])$,*

$$\lim_{N \rightarrow \infty} \frac{1}{\log(2 + N)} \sum_{n=1}^N \frac{\langle e_n, f(H)e_n \rangle}{n} = \frac{\tau_0(f) + \tau_1(f)}{2}.$$

Proof. Given $\varepsilon > 0$, choose by polynomial approximation a polynomial p with $\|f - p\|_{C([0,5])} < \varepsilon$. Functional calculus gives

$$|\langle e_n, (f(H) - p(H))e_n \rangle| \leq \varepsilon, \quad |\tau_c(f) - \tau_c(p)| \leq \varepsilon.$$

The harmonic sum divided by $\log(2 + N)$ is bounded and converges to one. Lemma 2, followed by $\varepsilon \downarrow 0$, yields the claimed limit. $\square$

4.4 From diagonal averages to Dixmier measurability

Fix $f \in C_c(\mathbb{R})$ and restrict it to $[0, 5]$. The source paper records the W -modulated eigenvalue formula [1, Theorem 4.5]: for every bounded operator T and $W = M_{1/n}$,

$$\sum_{k=1}^N \lambda_k(TW) = \sum_{n=1}^N \frac{\langle e_n, T e_n \rangle}{n} + O_T(1).$$

Apply it to $T = f(H)$. Lemma 3 shows that the normalized eigenvalue sums converge to $(\tau_0(f) + \tau_1(f))/2$. By the standard eigenvalue criterion for Dixmier measurability [3, Theorem 9.1.2(c)], $f(H)W$ is Dixmier measurable. This proves part (ii) and Theorem 1.

5 Verification and novelty check

The proof has four separable obligations: the arithmetic block oscillation, finite harmonic mass of interface neighborhoods, polynomial locality, and the passage from diagonal to eigenvalue sums. The first three are proved directly above; the fourth is exactly the source’s modulated-operator theorem. The included script `code/check_dyadic_schrodinger.py` checks the finite block averages and the logarithmic prediction for the cubic polynomial $1 + t + t^2 + t^3$. It is a sanity check only, not part of the proof.

Before promotion, the run indexes were searched for arXiv:2202.03676 and the phrases “Dixmier measurable density of states Schrödinger”, “discrete Schrödinger DOS converse”, and close variants. No prior run packet was found. A bounded web/arXiv search on 9 August 2026 used the exact source sentence, the paper title and authors, and the same close variants. It found the source paper and general DOS/Dixmier-trace work but no later explicit answer. Most notably, Hekkelman’s 2025 thesis (arXiv:2506.21950) repeats the discrete Schrödinger case as unknown [2, Sec. 5.1]. This is strong status evidence but not an exhaustive novelty claim.

6 Limitations and human-review recommendation

The theorem resolves only the discrete half-line question. It does not touch the source’s continuous Euclidean conjecture for $-\Delta + M_V$ on $L_2(\mathbb{R}^d)$. It also uses the particular admissible weight $w_n = 1/n$ from the source example; no assertion is made for every radial weak- ℓ_1 weight.

Expert review is recommended. The decisive checks are:

1. that the source’s phrase “discrete Schrödinger type operator” uses the standard class $\Delta_D + M_v$ (the example lies in that class);
2. that Theorem 4.5 of the source applies to $f(H)W$ exactly as used; and
3. that polynomial locality compares the half-line diagonal with the bi-infinite bulk diagonal outside the stated fixed-width exceptional set.

References

- [1] N. Azamov, E. Hekkelman, E. McDonald, F. Sukochev, and D. Zanin, *An application of singular traces to crystals and percolation*, arXiv:2202.03676 (2022).
- [2] E.-M. Hekkelman, *Trace Formulas in Noncommutative Geometry*, Ph.D. thesis, arXiv:2506.21950 (2025).
- [3] S. Lord, F. Sukochev, and D. Zanin, *Singular Traces, Volume 1: Theory*, De Gruyter, 2021.